29.3 A 28nm 15.59μJ/Token Full-Digital Bitline-Transpose CIM-Based Sparse Transformer Accelerator with Pipeline/Parallel Reconfigurable Modes

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Transformer models have achieved state-of-the-art results in many fields, like natural language processing and computer vision, but their large number of matrix multiplications (MM) result in substantial data movement and computation, causing high latency and energy. In recent years, computing-in-memory (CIM) has been demonstrated as an efficient MM architecture, but a Transformer's attention mechanism of raises new challenges for CIM in both memory access and computation aspects (Fig. 29.3.1): 1a) Unlike conventional static MM with pre-trained weights, the attention layers introduce dynamic MM (QK^T, $A'V$), whose weights and inputs are both generated at runtime, leading to redundant off-chip memory access for intermediate data. 1b) A CIM pipeline architecture can mitigate the above problem, but produces a new challenge. Since the K generation direction does not match the conventional CIM write direction, the QK^T-pipeline needs a large transpose buffer with extra overhead. 2) Compared with fully connected (FC) layers, attention layers dominate a Transformer's computation and require >8b precision to maintain accuracy, so previous analog CIMs [1-2] with ≤8b precision support cannot be directly used. Reducing the amount of computation for attention layers is critical for efficiency improvement.

To overcome the challenges, we propose a CIM-based Transformer accelerator TranCIM, with the following features: 1) TranCIM connects its CIM engines through a reconfigurable streaming network (RSN) with dedicated modes for different layers in the Transformer. For attention layers, TranCIM is configured in a pipeline mode that directly streams the 1st-stage engine's outputs to the 2nd-stage engine. For FC layers, TranCIM is configured in a parallel mode with all engines working independently. 2) Unlike previous wordline-direction-feeding solutions [1-3], our CIM macro is designed with a bitline-transpose structure to align the directions of input feeding and weight writing. Thus, in the QK^T-pipeline mode, transposing K is realized without additional storage and buffer access. The full-digital CIM design supports INT16 for attention layers and INT8 for FC layers to avoid accuracy loss caused by analog CIM logic. 3) Based on recent sparse Transformer algorithms [4-5], attention layers can be trained with block-sparse patterns to reduce computation while maintaining high accuracy. We design a sparse attention scheduler (SAS) to dynamically configure the CIM workload for different sparse attention patterns, realizing a computation complexity reduction from $O(N^2)$ to $O(N)$, where N=Token Count.

Figure 29.3.2 shows the TranCIM accelerator architecture, which comprises 2 static engines (SEngine0/1), a dynamic engine (DEngine), an RSN, an SAS, buffers, a SIMD core, and a top controller. Each engine has 16 bitline-transpose-CIM (BLT-CIM) macros and an engine accumulator to add up CIM outputs. Compared with SEngine, DEngine has extra configurations for data source selection and output aggregation. TranCIM's parallel mode is similar to previous CIM accelerators [1-3]. All the 3 engines store FC layer weights and work in parallel. TranCIM has two pipeline modes for QK^T- and $A'V$-MM in attention layers. In the QK^T-pipeline, TranCIM configures SEngine0 and SEngine1 as the 1st stage to compute K and Q, and DEngine as the 2nd stage to compute A. DEngine's CIMs are configured as independent outputs instead of combining them. The SIMD core performs scaling and softmax on A to generate A'. The $A'V$-pipeline has a similar dataflow, except that the 2nd stage's input A' is loaded from the global buffer. SEngines are combined to compute V, while DEngine computes the attention layer output Att. In this paper, we will mainly use the QK^T-pipeline to explain the features of TranCIM.

Figure 29.3.3 illustrates the QK^T-pipeline mode with a global+local sparse attention pattern. The block sparsity size is b, which means every b tokens of the input sequence are processed together to compute a b×b block of the attention matrix A. RSN activates the weight bus and input bus to construct a two-stage pipeline between SEngine and DEngine. SEngine0 and SEngine1 respectively store the weight matrices W_K and W_Q. The b tokens are sent into the engines bit-serially, so it takes bp cycles to compute one block of Q and K (p=precision). This period of time is defined as a block cycle. DEngine receives Q, K blocks through RSN, and multiplies them on its CIMs ($a_{i,j}=q_i \cdot k_j^T$). Taking the $a_{0,0}$ computation as an example: In block cycle1, SEngine0 generates k_0, and writes it as the weight in DEngine's CIM0. In block cycle2, SEngine1 generates q_0, and feeds it as the input for DEngine's CIM0, thus computing $a_{0,0}$ at the same time. The time chart illustrates the first 8 block cycles. Compared with the conventional parallel mode, the proposed pipeline mode loads the input sequence X only once, and directly reuses Q, K on-chip, achieving 2.25× fewer off-chip memory accesses and 2.14× energy saving.

Figure 29.3.4 shows the BLT-CIM macro that consists of a CIM controller, a 16×256 SRAM-CIM array and 16 macro accumulators. The 3 engines have the same CIM macro architecture, except that DEngine's CIM controller receives dynamic CIM workload configurations from the SAS in pipeline mode. A 6T SRAM cell and a 4T NOR gate are used for bitwise weight storage and multiplication. Input bits are fed to each row through the bitline. We utilize the inherent negation logic in SRAM to realize I AND W with $\overline{I}$ NOR $\overline{W}$, saving 2 transistors per cell. Every 16 columns of the array have a macro accumulator with 4 stages: 1) Partial product addition, 2) bit-serial accumulation, 3) INT8/INT16 fusion and 4) quantization. The full-digital CIM activates all rows simultaneously for full array utilization, without accuracy loss caused by analog computing [1-2]. In the QK^T-pipeline example, computing $a_{0,0}=q_0k_0^T$ demands DEngine's CIM0 store k_0 in a transpose fashion, but k_0 is generated along the b direction. If weights are written vertically as in conventional CIMs, an extra transpose buffer having the same size as CIM0 is needed to store the whole k_0 and then transpose write k_0 to CIM0. We solve the problem by aligning the input feeding and weight writing directions. BLT-CIM places bitlines horizontally and uses bitlines instead of wordlines to receive inputs for computation. Therefore, k_0 can be directly written into CIM0 and then multiplied by q_0 without pipeline stalls, saving 1.26× area, 1.07× power and 3.67× the SRAM access energy for each CIM write.

Sparse Transformer algorithms can significantly reduce attention computation [4-5], but different sparse attention patterns lead to various $q_i \cdot k_j^T$ pairs in matrix A. Thus, in QK^T-pipeline mode, DEngine's CIM workload always changes between different block cycles, which is supported by the SAS (Fig. 29.3.5). The SAS comprises a sparse attention initializer, an attention matcher, a DEngine CIM manager, and a configuration generator. The top controller writes the binary attention pattern and weight block lifetime into the sparse attention initializer during mode configuration. The DEngine CIM manager maintains the CIM workload information: Input ID, Weight ID, and Current Lifetime. In pipeline mode, once the SEngine generates a new block, the sparse attention initializer searches the $q_i \cdot k_j^T$ pairs. Then, the attention matcher compares the attended blocks with the weight blocks in the DEngine, determining whether to use the new block for input feeding or weight writing. In the example, q_1 attends to k_0 and k_1, matching the two blocks in DEngine, so the manager sets q_1 as the input for CIM0 and CIM1. k_2 has no matched blocks, so it's set as the new weight for CIM2. The CIM's weight lifetime decreases if new inputs come. Based on the manager's updated information, the SAS generates configurations for the DEngine, controlling each CIM macro's data source and work state. The SAS reduces attention computation by assigning only dense and necessary workloads to the DEngine. In an example QK^T-MM computation with a token count of 1024-4096, the SAS achieves 4.9-to-16.93× speedup and 3.51-to-11.25× energy savings through global+local sparse attention.

Figure 29.3.6 shows the measurement results of the TranCIM, fabricated in 28nm CMOS. The chip can work at 0.6-1.0V, 80-240MHz. The highest energy efficiency is 5.1TOPS/W for INT16 and 20.5TOPS/W for INT8 at 0.65V, 125MHz. Experiments are conducted on two Transformer models, BERT-base and BERT-large, with the Natural Questions dataset. Considering the large model size, we include off-chip DDR3 memory access for complete-system evaluation. In comparison to the conventional parallel mode with full attention, TranCIM's pipeline mode and sparse attention contribute to 5.22× speedup and 7.99× energy savings on attention layers of BERT-base, obtaining 2.63× speedup and 2.39× on the entire model. Compared with the state-of-the-art digital CIM accelerator [3], TranCIM consumes only 15.59μJ/Token for BERT-base, achieving 12.08× chip-level energy reduction. TranCIM's die photo and summary table are shown in Fig. 29.3.7.

Acknowledgement:
This work was supported in part by NSFC Grant 61774094 and U19B2041, National Key R&D Program 2018YFB2202600, Beijing S&T Project Z191100007519016 and Beijing Innovation Center for Future Chip. The corresponding author of this paper is Shouyi Yin (yinsy@tsinghua.edu.cn).

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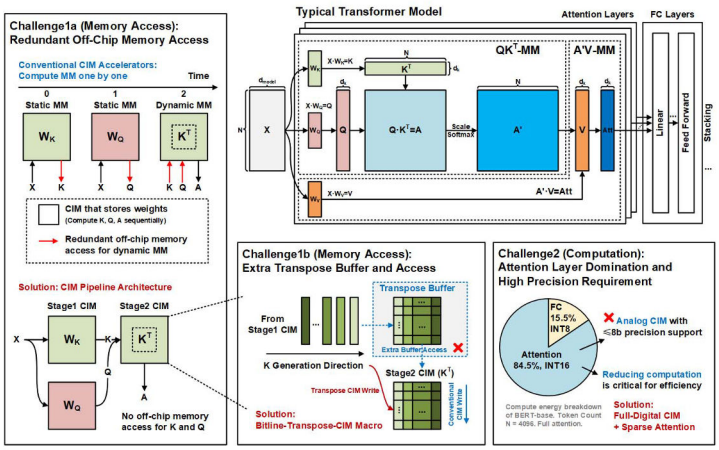


Figure 29.3.1: Transformer's attention mechanism raises three challenges for CIM accelerator design.

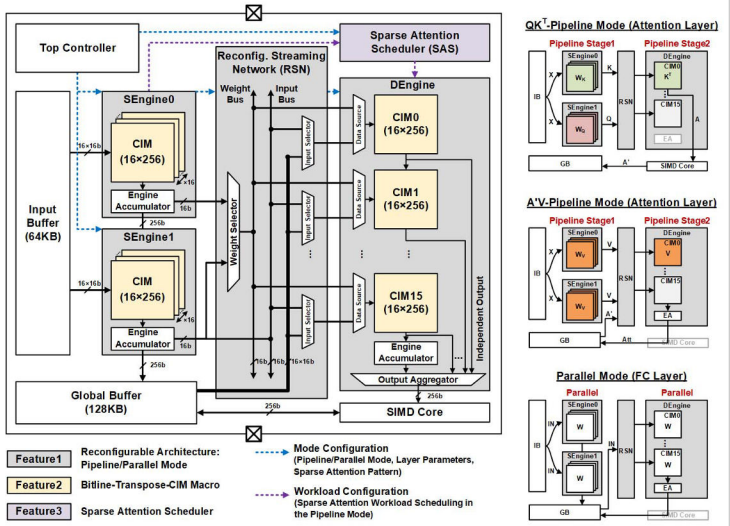


Figure 29.3.2: TranCIM's overall architecture, three main design features, and pipeline/parallel modes.

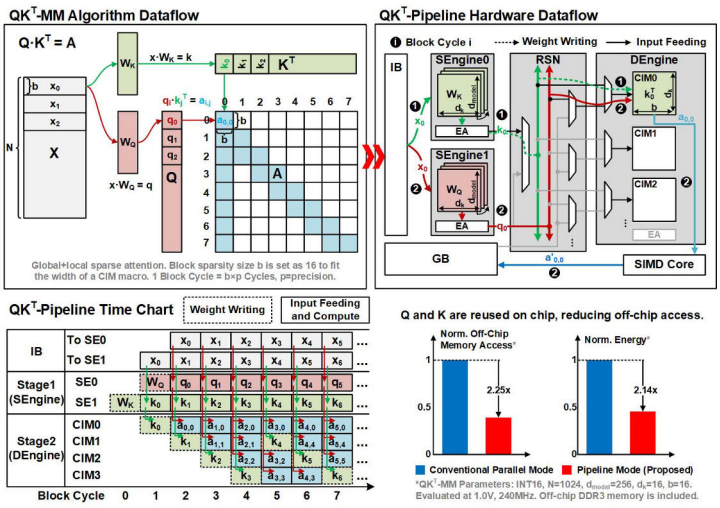


Figure 29.3.3: TranCIM's pipeline mode (QKT-pipeline as an example).

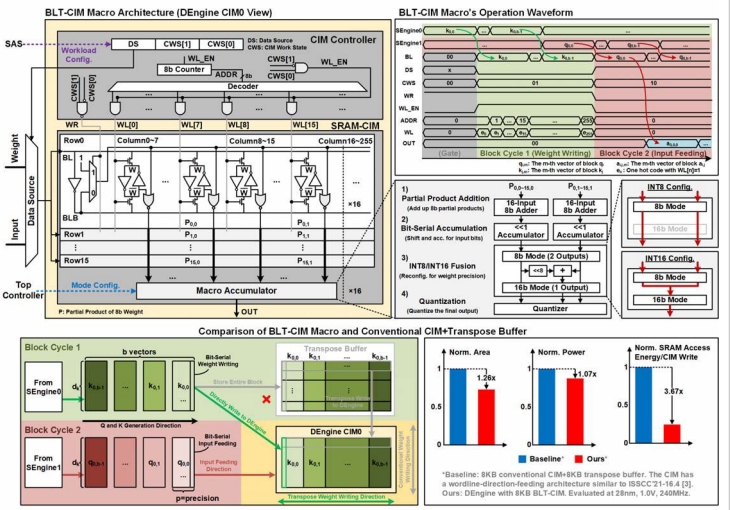


Figure 29.3.4: Bitline-Transpose-CIM (BLT-CIM) macro and comparison with conventional CIM+transpose buffer architecture.

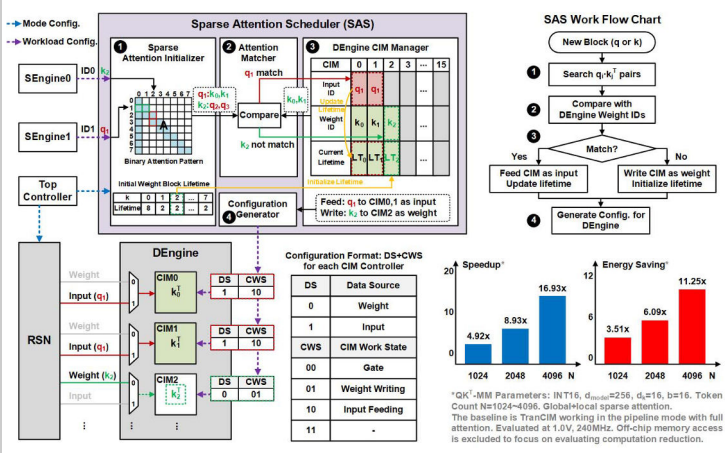


Figure 29.3.5: Sparse attention scheduler (SAS) and workload configuration for the DEngine in the pipeline mode.

Evaluation on Different Transformer Models

Model	BERT-base	BERT-large
Dataset	Natural Questions	
Data Precision	INT16 for attention layers, INT8 for FC layers	
Accuracy (LA/SA)	0.736/0.544	0.765/0.570
Accuracy Loss*1	-0.012/-0.008	-0.014/-0.011
Execution Time*2	1684ms (DDR3 included)	6646ms (DDR3 included)
Speedup*2,3	5.22x (attention layers) 2.63x (entire model)	4.27x (attention layers) 2.12x (entire model)
Energy Saving*2,3	7.99x (attention layers) 2.39x (entire model)	6.58x (attention layers) 1.87x (entire model)
Energy/Token*4	15.59μJ/Token	55.10μJ/Token

*1: Compared with accuracy at FP32. Use F1 score as the metric. Token Count N=4096. Global+local sparse attention.
*2: Off-chip DDR3 memory is included into latency and energy. Measured at 1.0V, 240MHz for high-performance evaluation.
*3: The baseline is TranCIM working in the parallel mode with full attention.
*4: Off-chip DDR3 memory is excluded to evaluate chip-level energy consumption. Measured at 0.65V, 125MHz for high-efficiency evaluation.

Voltage-Frequency Scaling
Frequency (MHz)
Power (mW)
Voltage (V)
Highest efficiency point 0.65V, 125MHz

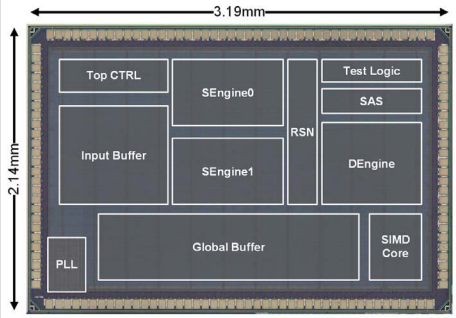
Comparison with State-of-the-art CIM Accelerators

		ISSCC'21-15.2 [1]	ISSCC'21-16.3 [2]	ISSCC'21-16.4 [3]	This Work
MAC Implementation		Analog CIM	Analog CIM	Digital CIM	Digital CIM
Data Precision		INT2/4/6/8	INT4/8	INT4/8/12/16	√ INT8/16
Work Mode		Parallel	Parallel	Parallel	√ Pipeline/Parallel
Technology		65nm	28nm	22nm	28nm
Supply Voltage (V)		0.62-1.0	0.7-0.9	0.72	0.6-1.0
Frequency (MHz)		25-100	138.9-250	500	80-240
Die Area (mm²)		12	0.772 (array)	0.202 (macro)	6.83
Power (mW)		18.6-84.1	-	-	27.04-118.21
Peak Performance*1 (TOPS)	INT8	0.013	0.213	0.917	1.48*4
	INT16	N/A	N/A	0.243	0.37*4
Energy Efficiency*1 (TOPS/W)	INT8	2.75 (INT8/4)	7.29*2	11.85*2	20.5*5
	INT16	N/A	N/A	3.14*2,3	5.1*5
Energy Consumption*1 on BERT-base (μJ/Token)		N/A (w/o INT16 support)	N/A (w/o INT16 support)	188.33 (12.08x)*3	15.59 (1x)*5

*1: Off-chip DDR3 memory is excluded for chip-level evaluation. One operation (OP) represents one multiplication or one addition.
*2: On-chip buffers are included for fair comparison. Buffer capacity is 192KB, same as TranCIM.
*3: Prior accelerators can only work in the parallel mode with full attention.
*4: Measured at the highest performance point, 1.0V, 240MHz.
*5: Measured at the highest efficiency point, 0.65V, 125MHz. TranCIM has pipeline/parallel modes and sparse attention support.

Figure 29.3.6: Measurement results and comparison with state-of-the-art CIM accelerators.

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Technology	TSMC 28nm CMOS	
Die Area	2.14mm×3.19mm	
Supply Voltage	0.6–1.0V	
Frequency	80–240MHz	
Buffer Size	192KB	
CIM Size	24KB	
Data Precision	INT8, INT16	
Work Mode	Pipeline/Parallel	
Power	27.04mW @0.6V, 80MHz	
	118.21mW @1.0V, 240MHz	
Peak Performance[1]	INT8	1.48TOPS
	INT16	0.37TOPS
Energy Efficiency[1,2]	INT8	12.5-20.5TOPS/W
	INT16	3.1-5.1TOPS/W
Energy/Token[2]	BERT-base	15.59μJ/Token
	BERT-large	55.10μJ/Token

*1: Highest performance (lowest efficiency) point, 1.0V, 240MHz.
*2: Highest efficiency point, 0.65V, 125MHz. Chip-level energy efficiency includes all on-chip components. Token Count N=4096. Global+local sparse attention.

Figure 29.3.7: Die micrograph and summary table.

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